

To what extent does sympathetic physiological reactivity predict attentional bias towards threatening stimuli in adults with varying levels of trait anxiety?

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Proposal Declaration

I, Diana Mamonova, declare that this proposal has been composed solely by myself and that it has not been submitted, in whole or in part, in any previous application for a degree. Except where stated otherwise, by reference or acknowledgement, the work presented is entirely my own and has been generated by me as the result of my own original research.

If generative artificial intelligence has been used, I have acknowledged its use in accordance with the University's current academic-integrity requirements.

I remain responsible for checking all claims and sources, ensuring that the final submission is in my own words and reflects my own understanding.

Signed: _____Diana Mamonova_____

Date: _____14 September 2026_____

To what extent does sympathetic physiological reactivity predict attentional bias towards threatening stimuli in adults with varying levels of trait anxiety?

Background Literature Review

Anxiety involves both cognitive and physiological processes. Cognitive theories suggest that people with anxiety pay more attention to threatening information than neutral information, which may increase feelings of danger and help maintain anxiety symptoms (Beck & Clark, 1997). This tendency is known as attentional bias towards threat. It refers to a greater focus on threatening stimuli, difficulty shifting attention away from them, or avoiding them altogether (Cisler & Koster, 2010). Research has consistently found that anxious individuals show greater attentional bias towards threat than non-anxious individuals, although findings vary across different tasks and studies (Bar-Haim et al., 2007).

Anxiety is also associated with changes in the body's physiological response to threat. The autonomic nervous system helps regulate these responses, particularly through activation of the sympathetic nervous system. One way to measure sympathetic activity is through skin conductance response (SCR), which reflects changes in sweat gland activity when people encounter emotionally significant or threatening stimuli (Boucsein, 2012; Dawson et al., 2017). SCR is widely used in psychophysiological research as a non-invasive measure of physiological reactivity.

The present proposal uses the term sympathetic physiological reactivity rather than the broader construct physiological arousal. This distinction addresses the fact that SCR and heart-rate variability are not interchangeable. Whereas SCR is directly driven by sympathetic sudomotor activity, commonly used short-term heart-rate variability indices are strongly influenced by parasympathetic regulation. HRV is consequently excluded to maintain a single, theoretically coherent predictor.

From a cognitive neuroscience perspective, threat processing involves several brain regions. The amygdala plays an important role in detecting threats, while areas of the prefrontal cortex and anterior cingulate cortex are involved in attention and emotional regulation (Bishop, 2007). These systems help explain why physiological responses and attentional processes may be closely related. Although the present study will not directly measure brain activity, it aims to examine whether sympathetic physiological responses to threat are associated with attentional bias towards threatening information.

Prior Findings

The dot-probe task is widely used to operationalise threat-related attention. A threat-neutral stimulus pair is displayed, followed by a probe in the location of either stimulus. Faster responses on threat-congruent than threat-incongruent trials are interpreted as

relatively greater allocation of attention towards threat (MacLeod et al., 1986). The conventional bias score is calculated as mean reaction time on threat-incongruent trials minus mean reaction time on threat-congruent trials, so a positive value indicates bias towards threat.

However, studies examining attentional bias towards threat have produced inconsistent findings. Trial-level research suggests that vigilance, avoidance and disengagement patterns can be obscured when performance is reduced to one mean score (Evans et al., 2020). Recent work has also questioned whether trait anxiety consistently predicts dot-probe bias and has highlighted reliability limitations (Williams et al., 2024). These concerns do not make the paradigm invalid, but require careful task design, adequate trial numbers, transparent scoring and cautious interpretation.

Research has separately associated anxiety with altered threat attention and autonomic responding, but fewer studies have directly tested whether stimulus-evoked sympathetic reactivity predicts attentional bias and whether this association varies across trait anxiety. Measuring trait anxiety continuously retains individual-difference variability that would be lost by recruiting only high scorers. It also permits a moderation analysis rather than treating anxiety solely as an eligibility category.

Theory

The proposal is informed by cognitive models in which threat-related schemas increase prioritisation of danger cues (Beck & Clark, 1997), and by Attentional Control Theory, which argues that anxiety disrupts goal-directed control and increases the influence of stimulus-driven processing (Eysenck et al., 2007). Bringing these ideas together suggests the following mechanism: people who are more sensitive to threat tend to show a stronger sympathetic nervous system response when they encounter threat, and this stronger response is linked to greater attention towards threatening information. Trait anxiety may strengthen this link because people with higher trait anxiety are more likely to interpret ambiguous or negative cues as threatening. However, this is a theoretical and statistical prediction, not a claim that skin conductance response (SCR) directly causes attentional bias.

Rationale and Hypotheses

The study addresses the question: To what extent does sympathetic physiological reactivity, measured by SCR, predict attentional bias towards threatening stimuli, and is this relationship moderated by trait anxiety in adults? This study is important because it examines physiological and cognitive processes together rather than separately.

H1: Greater threat-specific SCR will significantly predict a more positive dot-probe attentional-bias score.

H2: Trait anxiety will moderate the relationship between threat-specific SCR and attentional bias, such that the positive association will be stronger at higher trait-anxiety levels.

The corresponding null hypotheses are that threat-specific SCR will not predict attentional-bias scores and that the SCR $\times$ trait-anxiety interaction will not explain significant additional variance.

Aims and Objectives

The primary aim is to determine whether sympathetic reactivity to threat predicts attentional bias and whether trait anxiety moderates that association. The objectives are to: (1) quantify threat-specific SCR; (2) produce a valid reaction-time attentional-bias score from a facial dot-probe task; (3) measure trait anxiety continuously using the trait scale of the State-Trait Anxiety Inventory; and (4) test the predictor, moderator and interaction within a prespecified regression model.

Research Design

Design

A quantitative, cross-sectional, laboratory-based correlational design will be used. The predictor will be threat-specific SCR, operationalised as the mean event-related SCR amplitude for threatening-face trials minus the mean amplitude for neutral control trials. The outcome will be the dot-probe attentional-bias score. Trait anxiety will be the continuous moderator. Because variables are measured rather than manipulated, the design can establish statistical prediction and association but not causation.

Materials

Electrodermal activity will be recorded using calibrated skin-conductance equipment with disposable electrodes placed on the palmar surface of the non-dominant hand. A brief resting baseline will precede the task. Event markers will synchronise stimulus onset and physiological recording. SCR scoring windows, minimum response threshold and treatment of non-responses will be specified before data collection and applied consistently in accordance with electrodermal measurement guidance (Boucsein, 2012).

The computerised dot-probe task will present validated angry and neutral facial expressions from the same actors, counterbalanced across screen position and probe congruency. Each trial will include fixation, a face pair, and a probe requiring a left or right key response. Accuracy and reaction time will be recorded. Incorrect responses, anticipatory responses and implausibly slow responses will be removed using prespecified criteria. The trait subscale of the State-Trait Anxiety Inventory will measure relatively stable anxiety proneness (Spielberger et al., 1983). A demographic and health questionnaire (to follow), will record age, corrected vision, relevant medication, caffeine, nicotine, recent exercise and sleep because these may affect autonomic measurement.

Participants

Approximately 80 adults aged 18–65 will be recruited from the university participant pool and community advertisements. The final sample size should be confirmed using a power analysis carried out before the study. This analysis should be based on an expected effect size from previous research, with a significance level of .05 and statistical power of .80.

Eligible participants must be able to provide informed consent, understand written English and have normal or corrected-to-normal vision. Exclusion criteria will include a self-reported neurological condition, current psychosis, significant uncorrected visual impairment, or medication known to substantially affect autonomic functioning. Exclusions will be applied cautiously and transparently, and participants

will not be asked to stop prescribed medication. Participants will be informed that they have the right to withdraw from the study at any time without penalty and without providing a reason. Recruitment across the full trait-anxiety range avoids range restriction and supports the moderation question.

Data Analysis Plan

Data will be analysed in IBM SPSS Statistics or equivalent software. Preparation will include accuracy checks, missing-data assessment, preregistered reaction-time exclusions, inspection of SCR artefacts and calculation of threat-specific SCR and attentional-bias scores. Internal consistency of the STAI trait scale will be reported. Descriptive statistics and zero-order correlations will summarise the variables.

The primary analysis will be hierarchical multiple regression. Mean-centred threat-specific SCR and trait anxiety will be entered as main effects, followed by their product term. H1 will be evaluated from the SCR regression coefficient. H2 will be evaluated from the interaction coefficient and change in explained variance after adding the interaction. A significant interaction will be probed using simple slopes at low, mean and high trait-anxiety values, with confidence intervals reported. Regression assumptions will be assessed through residual plots, normality of residuals, homoscedasticity, linearity, influence statistics and variance-inflation factors. If assumptions are violated, additional analyses will be considered to check the robustness of findings.

This analysis is suitable because the study examines whether a continuous physiological measure predicts a continuous behavioural outcome, and whether this relationship changes depending on another continuous variable. Statistical significance will be assessed at $p < .05$, but greater emphasis will be placed on effect sizes and 95% confidence intervals. The third hypothesis from the earlier outline, which suggested splitting participants into high- and low-anxiety groups, was removed because dividing STAI scores into categories would reduce the amount of information available and does not fit with the continuous moderation approach.

Procedure

After ethics approval, prospective participants will receive an information sheet and provide informed consent. They will be reminded that participation is voluntary and that withdrawal is possible without disadvantage. Participants will complete the demographic and health questions and STAI trait scale. The electrode sites will be cleaned, sensors attached and participants will be explained that the equipment records sweat-related electrical conductance, not thoughts or diagnoses.

Following a seated acclimatisation and baseline period, participants will complete practice trials and then the dot-probe task while electrodermal activity is recorded continuously. Standardised instructions, room conditions and task timing will reduce avoidable measurement variability. On completion, sensors will be removed, participants will be asked about discomfort, and a written debrief will explain the aims without claiming that individual scores diagnose anxiety. The session is expected to last circa 35 - 45 minutes.

Ethics

Approval will be sought from the University of Essex Online, Department of Psychology and Criminology, before recruitment or data collection. The study will follow the British Psychological Society Code of Human Research Ethics (British Psychological Society, 2021). Although procedures are non-invasive, threatening faces and anxiety questions may cause temporary discomfort. The information sheet will describe this foreseeable risk without exaggeration, and participants may pause or discontinue the task. Distress protocol will be in place, participants will be provided with a debrief and informed of appropriate wellbeing support. Immediate risk disclosures will be managed under the limits to confidentiality stated in the approved materials.

Consent must be fully informed, given voluntarily, and recorded. Participants' data will be protected by using participant ID codes instead of names, and consent forms will be stored separately from the research data.

Access will be restricted, retention and deletion periods will follow university policy, and reporting will use aggregate results. Participants will be told how withdrawal of identifiable data operates and the point after which anonymised data may no longer be removable. Because anxiety is not a protected clinical diagnosis within this study, recruitment materials will avoid stigmatising language and will not imply treatment or clinical assessment. Skin preparation and sensor use will follow hygiene procedures, and participants may decline electrode placement.

Limitations

The cross-sectional correlational design cannot demonstrate that sympathetic responding causes attentional bias; a shared process or unmeasured factor could explain an association. SCR is sensitive to temperature, movement, hydration, medication, caffeine and individual differences in responsivity. Standardisation and measured covariates can reduce but not eliminate these influences. The dot-probe task has documented reliability concerns, and a mean bias score may conceal dynamic vigilance and avoidance patterns. Adequate trials, prespecified exclusions and transparent reporting will mitigate, rather than remove, this limitation.

Using static facial expressions may not fully reflect how people respond to emotions in real-life situations, and recruiting mainly university students and volunteers may limit how well the findings apply to wider populations. A sample size based on a medium interaction effect may not detect a smaller effect, so the power analysis and confidence intervals will be important. Finally, measuring SCR while participants complete the task can show that the two occur at the same time, but it cannot show which causes the other. Future research could use eye tracking, more detailed measures of attentional bias, repeated testing, or experimental changes in arousal to examine the proposed mechanism more directly.

Timeline

Table 1

Proposed Research Timeline

Work package	Main activities	Indicative timing
1. Finalise research protocol	Finalise measures, procedure, participant materials, and analysis plan.	Weeks 1–2
2. Ethics review	Submit the ethics application and address any requested revisions.	Weeks 3–5
3. Recruitment and data collection	Recruit eligible participants and complete data collection after ethical approval.	Weeks 6–7
4. Data preparation	Preprocess collected data, manage artefacts, and screen the dataset.	Weeks 8–11
5. Data analysis	Conduct the prespecified statistical analyses and summarise the findings.	Weeks 12–13
6. Write up	Complete the results, discussion, and remaining project sections.	Week 14
7. Revision and submission	Proofread, check APA formatting and references,	Weeks 15–16

	and prepare the final submission	
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Appendices

Appendix A

Participant Information Sheet

The Relationship Between Physiological Response to Threat, Attention to Threatening Information, and Trait Anxiety

Invitation to Take Part

You are being invited to take part in a research study investigating how people respond to threatening information. Before you decide whether to participate, please read the following information carefully. It explains why the research is being conducted and what your participation would involve.

Please ask the researcher if anything is unclear or if you would like more information.

What is the Purpose of the Study?

The aim of this study is to explore whether physiological responses to threat are associated with attentional bias towards threatening information. The study will also investigate whether this relationship varies depending on levels of trait anxiety.

Trait anxiety refers to a person's general tendency to experience anxiety across situations. Physiological responses will be measured using skin conductance response (SCR), which reflects changes in sweating controlled by the nervous system.

Why Have I Been Invited?

You have been invited because you are aged 18 years or over and meet the eligibility criteria for participation.

We aim to recruit participants with a wide range of anxiety levels to better understand how anxiety may influence responses to threatening information.

Some people may not be eligible to participate. For example, individuals with certain neurological conditions, significant visual impairments, or those taking medication that substantially affects physiological responses may be unable to take part. The researcher will explain these requirements before participation.

Do I Have to Take Part?

No. Participation is entirely voluntary.

If you choose to take part, you will be asked to provide informed consent before the study begins. You are free to withdraw at any time without giving a reason and without penalty. If you withdraw during the study, any data collected up to that point can be removed if you request this.

Choosing not to participate or deciding to withdraw will not affect you in any way.

What Will Happen If I Take Part?

If you decide to take part, you will be asked to:

1. Read this information sheet and provide informed consent.
2. Complete a questionnaire measuring trait anxiety.
3. Complete a computer-based attention task involving images of faces displaying different emotional expressions.
4. Wear sensors on your fingers that measure skin conductance response (SCR) during the task.

The study is expected to take approximately **30-45 minutes**.

Are There Any Risks or Discomforts?

Some participants may find images showing angry facial expressions mildly uncomfortable. However, these images are commonly used in psychological research and are not expected to cause significant distress.

You may take a break or stop participating at any time if you feel uncomfortable.

The study does not involve any invasive procedures. SCR sensors only measure small changes in skin conductance and do not deliver any electrical stimulation.

What Are the Possible Benefits?

You may not receive any direct benefit from taking part.

However, your participation will help researchers better understand how physiological responses and anxiety relate to attention to potentially threatening information. This knowledge may contribute to future research on anxiety and emotional processing.

What Will Happen to My Data?

All information collected during the study will be kept confidential.

Your data will be assigned a participant identification number rather than your name. Consent forms will be stored separately from research data so that your responses cannot be directly linked to your identity.

Only the research team will have access to the data.

The data will be stored securely in accordance with university data protection policies and relevant data protection legislation, including the UK General Data Protection Regulation (UK GDPR).

What Will Happen to the Results?

The results of the study may be included in a university dissertation, academic presentations, or scientific publications.

Any reports arising from the study will present findings in summary form only. Individual participants will not be identifiable.

Who Has Reviewed This Study?

This research has been reviewed and approved by the relevant University Research Ethics Committee before data collection begins.

What If I Have Questions or Concerns?

If you have any questions about the study, please contact:

Researcher:

Diana Mamonova

07512345678

Diana.mamonova@essex.co.uk

Appendix B

Participant Consent Form

The Relationship Between Physiological Response to Threat, Attention to Threatening Information, and Trait Anxiety

Researcher

Diana Mamonova

Supervisor

Mrs Supervisor

Please read each statement carefully and tick the box if you agree.

☐ I confirm that I have read and understood the Participant Information Sheet for this study.

☐ I have had the opportunity to ask questions about the study and have received satisfactory answers.

☐ I understand that my participation is voluntary.

☐ I understand that I am free to withdraw from the study at any time without giving a reason and without penalty.

☐ I understand that the information I provide will be treated confidentially.

☐ I understand that my data will be identified using a participant code rather than my name.

☐ I understand that the study involves completing questionnaires, a computer-based attention task, and the collection of skin conductance response (SCR) data through sensors attached to my fingers.

☐ I understand that some facial images used in the study may depict angry or threatening expressions, which some people may find mildly uncomfortable.

☐ I understand that anonymised findings may be used in research reports, dissertations, conference presentations, or academic publications.

☐ I agree to take part in this study.

I understand that my personal data will be processed in accordance with UK GDPR and university data protection policies. Any identifying information will be stored securely and separately from research data, and only authorised members of the research team will have access to it.

☐ I consent to the processing of my data as described above.

Participant

Name: _____

Signature: _____

Date: _____

Researcher

Name: _____

Signature: _____

Date: _____

Appendix C

Debrief Form

Thank you for taking part in this study.

The purpose of this research is to investigate whether physiological responses to threat are related to attentional bias towards threatening information, and whether this relationship is influenced by trait anxiety.

Attentional bias refers to the tendency to pay more attention to certain types of information than others. In this study, we are interested in attention towards threatening facial expressions.

To explore this, participants completed:

- A questionnaire measuring trait anxiety.
- A computer-based task involving emotional facial expressions.
- Skin conductance response (SCR) recording during the task.

SCR is a commonly used measure of sympathetic nervous system activity and reflects changes in skin conductance associated with emotional arousal.

Research Question

The study aims to examine whether individuals who show stronger physiological responses to threat also show greater attentional bias towards threatening information, and whether this relationship differs according to levels of trait anxiety.

Why Was Full Information Not Provided Before Participation?

To ensure that participants respond as naturally as possible, the precise hypotheses were not fully explained before participation. Providing detailed information beforehand could have influenced responses and affected the validity of the findings.

Now that your participation is complete, the full purpose of the research has been explained.

Right to Withdraw Your Data

You have the right to withdraw your data from the study without penalty.

If you wish to withdraw your data, please contact the researcher within 14 days.

After this date, the data will have been anonymised and incorporated into the dataset, making individual data removal impossible.

Confidentiality

All information collected during this study will remain confidential.

Your data will be identified using a participant code rather than your name. Consent forms will be stored separately from research data. Findings will be reported only in summary form and no individual participant will be identifiable.

Questions or Concerns

If you have questions regarding the study, please contact:

Researcher

Diana Mamonova

07512345678

Diana.mamonova@essex.co.uk

Appendix D

Distress Protocol

Purpose

Although this study is considered low risk, some participants may experience mild discomfort when viewing angry or threatening facial expressions or when reflecting on anxiety-related questions.

The purpose of this protocol is to ensure that any participant who becomes distressed is supported appropriately.

Recognising Distress

Possible signs of distress may include:

- Visible upset or crying.
- Signs of anxiety or panic.
- Difficulty continuing the study.
- Verbal expressions of discomfort.
- Requests to stop participation.
- Withdrawal from interaction.

Actions During the Study

If a participant appears distressed:

1. The task will be paused immediately.
2. The researcher will check on the participant's wellbeing and ask whether they would like to continue, take a break, or stop participation.
3. The participant will be reminded that participation is voluntary and that they may withdraw at any time without giving a reason.
4. No pressure will be placed on the participant to continue.

If the Participant Chooses to Withdraw

If the participant decides to stop:

- The study session will end immediately.
- The participant will be thanked for their time.
- Information about support services will be provided.

- The participant will be informed of their right to withdraw their data within the specified withdrawal period.

Post-Study Support Information

If participation raises concerns or causes emotional discomfort, participants may wish to contact:

NHS Services

- NHS 111 (non-emergency support)
- GP or primary healthcare provider

University Support Services

- University Counselling Service
- Student Wellbeing Service
- Personal Tutor or Academic Advisor

Independent Support

Samaritans

- Telephone: 116 123
- Available 24 hours a day
- Website: www.samaritans.org

Mind

- Website: www.mind.org.uk

Researcher Responsibilities

The researcher should:

- Remain calm, supportive and non-judgemental.
- Respect the participant's decision regarding continuation or withdrawal.
- Maintain participant confidentiality.
- Record any adverse events according to university ethical procedures.
- Inform the supervisor of any significant adverse event as soon as possible.

Risk Assessment Summary

The study is considered minimal risk because:

- Participants view standardised facial expression stimuli commonly used in psychological research.
- SCR recording is non-invasive and does not involve electrical stimulation.
- Participants can pause or withdraw at any time without penalty.
- Appropriate support information is provided.